MAT 314 HW07

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Due 04/29

Problem. 1.1. Let R be an integral domain that contains a field F as a subring and that is finite-dimensional when viewed as a vector space over F. Prove that R is a field.

Solution. We show that any nonzero $r \in R$ has an inverse. If n is the dimension of the F-vector space R, then $1, r, \ldots, r^n$ is linearly dependent, satisfying some relation $c_0 + c_1 r + \cdots + c_n r^n = 0$. Let $k \geq 0$ be the smallest integer for which $c_k \neq 0$; obviously $k < n$. Since R is an integral domain we can deduce that

$$c_k + c_{k+1} r + \cdots + c_n r^{n-k} = 0 \iff r \cdot (-c_k^{-1})(c_{k+1} + c_{k+2} r + \cdots + c_n r^{n-k-1}) = 1$$

so that r has an inverse (this exact manipulation is also used in the following problem). $\square$

Problem. 2.2. Let $f(x) = x^n - a_{n-1}x^{n-1} + \cdots + (-1)^n a_0$ be an irreducible polynomial over F, and let α be a root of f in the extension field K. Determine the element α^{-1} explicitly in terms of α and the coefficients a_i.

Solution. Since f is irreducible over F, a_0 must be nonzero. Then, $f(\alpha) = 0$ so

$$-\frac{(-1)^n}{a_0}(f(\alpha) - (-1)^n a_0) = 1.$$

so the inverse of α is $-(-1)^n a_0^{-1} \cdot (\alpha^{n-1} - a_{n-1}\alpha^{n-2} + \cdots + (-1)^{n-1} a_1)$, factoring out α from $f(\alpha) - (-1)^n a_0$. $\square$

Problem. 3.1. Let F be a field, and let α be an element that generates a field extension of F of degree 5. Prove that α^2 generates the same extension.

Solution. Obviously $\alpha^2 \in F(\alpha)$, so $F(\alpha^2) \subseteq F(\alpha)$. On the other hand,

$$[F(\alpha) : F(\alpha^2)][F(\alpha^2) : F] = 5$$

and α is a root of the quadratic polynomial $p(X) = X^2 - \alpha^2 \in F(\alpha^2)[X]$, so the degree of $F(\alpha)$ over $F(\alpha^2)$ is either 1 or 2; being a divisor of 5, the degree must be 1, and so $F(\alpha) = F(\alpha^2)$. $\square$

Problem. 8.1. Prove that every finite extension of a finite field has a primitive element.

Solution. Let K be a finite extension of the finite field $\mathbb{F}_q$, with $q = p^m$ and $|K| = p^n$ for prime p and positive integers m, n. Then, if α generates the cyclic group $K^\times$, we can see that $\mathbb{F}_q(\alpha)$ on one hand contains K (since it contains $K^\times$, of course) but on the other hand is contained in K because $\mathbb{F}_q \subseteq K$ and $\alpha \in K$, so α is a primitive element for the extension. $\square$

Problem. 10.1. Prove that the subset of $\mathbb{C}$ consisting of the algebraic numbers is algebraically closed.

Solution. Let $\overline{\mathbb{Q}}$ be the set of algebraic numbers and let $p(x) = x^n + a_{n-1}x^{n-1} + \cdots + a_0$ be a polynomial with coefficients in $\overline{\mathbb{Q}}$. Then, p has coefficients in the finite extension $K = \mathbb{Q}(a_0, \ldots, a_{n-1})$ of $\mathbb{Q}$. We wish to show that the roots of p are also algebraic numbers, i.e. are contained in a finite extension of $\mathbb{Q}$. If β is any root of p, then $K(\beta) = \mathbb{Q}(a_0, \ldots, a_{n-1}, \beta)$ is spanned by $\{v_i \beta^j : 0 \leq i < m,\ 0 \leq \beta < n\}$ over $\mathbb{Q}$ where $\{v_1, \ldots, v_m\}$ is a basis for K. Thus $K(\beta)$ is a finite extension of $\mathbb{Q}$, so β is an algebraic number, hence any root of a polynomial with algebraic number coefficients is also algebraic. $\square$